

OpenClaw AI Pet Tutorial

Before starting, please complete [01-Peripheral hardware configuration], [02-openclaw API-KEY configuration], [03-openclaw switching model], [04-OpenClaw prompt words], [05-AI voice interaction configuration], [06-Three dialogue scheme configuration tests] in the `Preparation before use` directory. If you are currently only using TUI/WhatsApp, you can skip 05 first. This article assumes that the basic environment has been prepared and only the AI pet project itself will be expanded.

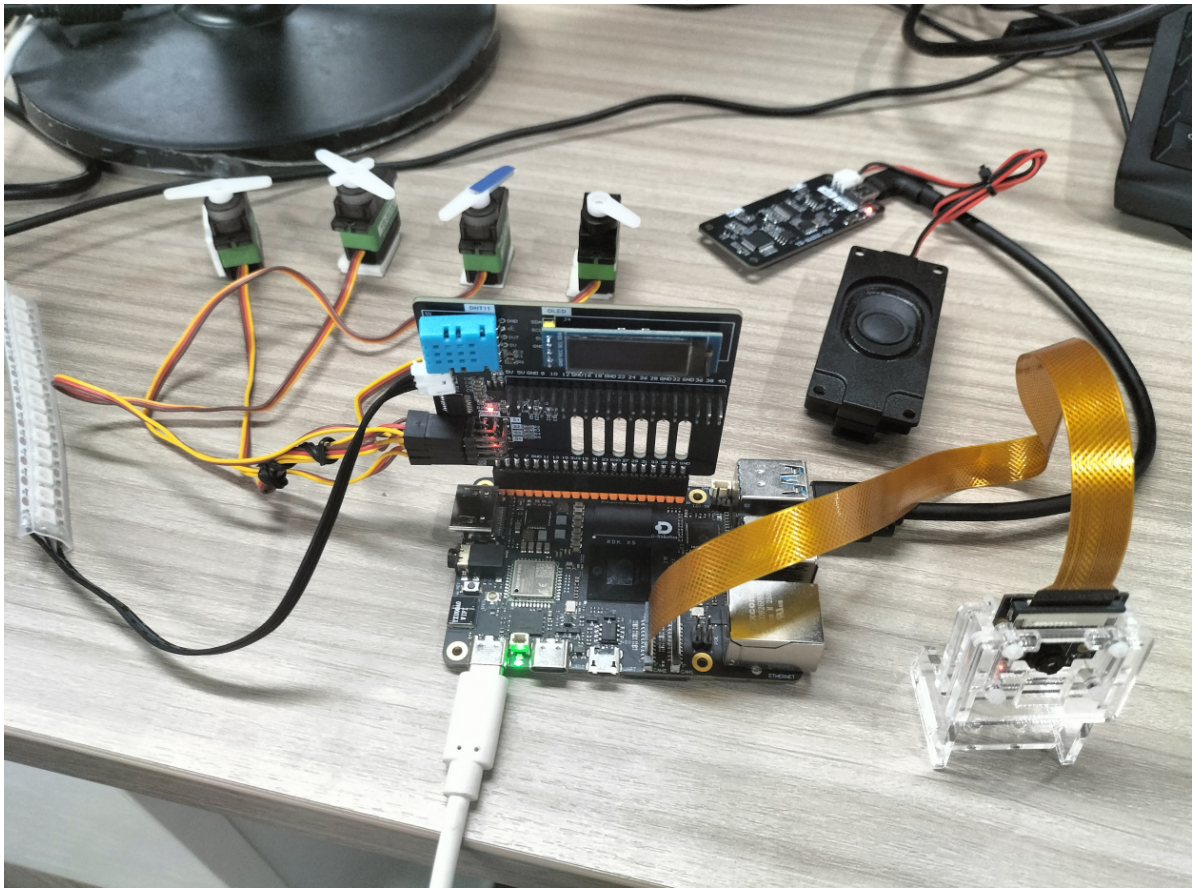
1. Tutorial Objectives

This article demonstrates the interactive capabilities of OpenClaw as a desktop AI pet through a combination of "action + lighting + screen + dialogue". After completing this tutorial, readers will be able to:

- Learn how to use servos, RGB and OLED to make an AI pet with emotional feedback
- Converse with OpenClaw through WhatsApp, TUI, and voice code
- Complete four project scenarios: greeting, emotional expression, companion Q&A, and sleep wake-up

2. Hardware preparation

- OpenClaw main control board
- Servo `[recommended at least 1]`
- RGB light strip/RGB lamp bead module
- OLED display
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software preparation

Code entry that will be used in the tutorial

- Servo control command: `/home/sunrise/hardware_hal_demo/cli/servo`
- Voice entrance: `/home/sunrise/voice_to_openc1aw/src/main.py`

This project does not have a fixed set of "pet scripts", but lets OpenClaw orchestrate several basic peripheral actions:

- The servo is responsible for nodding, shaking, and swinging
- RGB is responsible for expressing mood and status
- OLED is responsible for displaying names, emoticons or short text
- Voice dialogue is responsible for making the pet more like an interactive character

4. Test the basic capabilities of the project

Before starting the current project, please complete a basic dialogue self-test of `openc1aw tui` ; if you have not done so yet, please read [06-Three dialogue scheme configuration tests] first. After passing, you can enter the special test of the AI pet project.

Enter OpenClaw TUI:

```
openc1aw tui
```

```
sunrise@ubuntu:~/openclaw$ openclaw tui
♥ OpenClaw 2026.3.24 (cff6dc9) - You had me at 'openclaw gateway start.'

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main

hello

你好呀，主人！🐾 我是小亚，随时待命。

有什么需要我帮忙的吗？
gateway connected | idle
agent main | session main (openclaw-tui) | deepseek/deepseek-v4-pro | tokens 12k/1.0m (1%)
```

Then test several basic abilities:

- Let the servo nod
- Change the light to a pink breathing light
- Display "Hello" on OLED

5. Three options for talking to OpenClaw

Solution	Suitable scenario	Advantages	Suggested positioning
WhatsApp	Remote interaction, classroom demonstration	Suitable for demonstration of "remote pet raising"	Showcase character interaction capabilities
TUI	Local debugging, action verification	The easiest way to check whether actions and feedback match	Run through the character action logic first
Voice code	The ultimate interactive experience	The most like desktop pet that can really chat	As the final demonstration solution

If the AI voice module + speaker is optional, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

- Modify whether to open broadcast parameters

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter is to control whether to broadcast. After modification, save it and run the following program on the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp solution

For WhatsApp access and robot creation steps, please refer directly to [05-Configuration Test of Three Dialogue Schemes]. Once created, you can remotely interact with the AI pet in WhatsApp, for example:

- Whether there is dog food at the moment when shooting, if there is, the OLED will display a smiling expression, the servo will dance, and the OLED will turn on the ambient light. If there is no dog food, the OLED will display a sad expression.

20:31



< 2



+86 153 3885 7526 (你)

给自己发消息

Take a photo to see if there is dog food. If yes, display smile expression on OLED,make servo dance,and light up ambient lights on OLED. If no dog food,display sad expression on OLED.

20:31 ✓✓



I can see the scene, but there's **no dog food** visible in the image. I can see a desk with various items like a pink mug, green fan, water bottle, and a curriculum schedule, but no dog food detected

20:31 ✓✓

Dog food check completed!

Result: No dog food detected in the scene



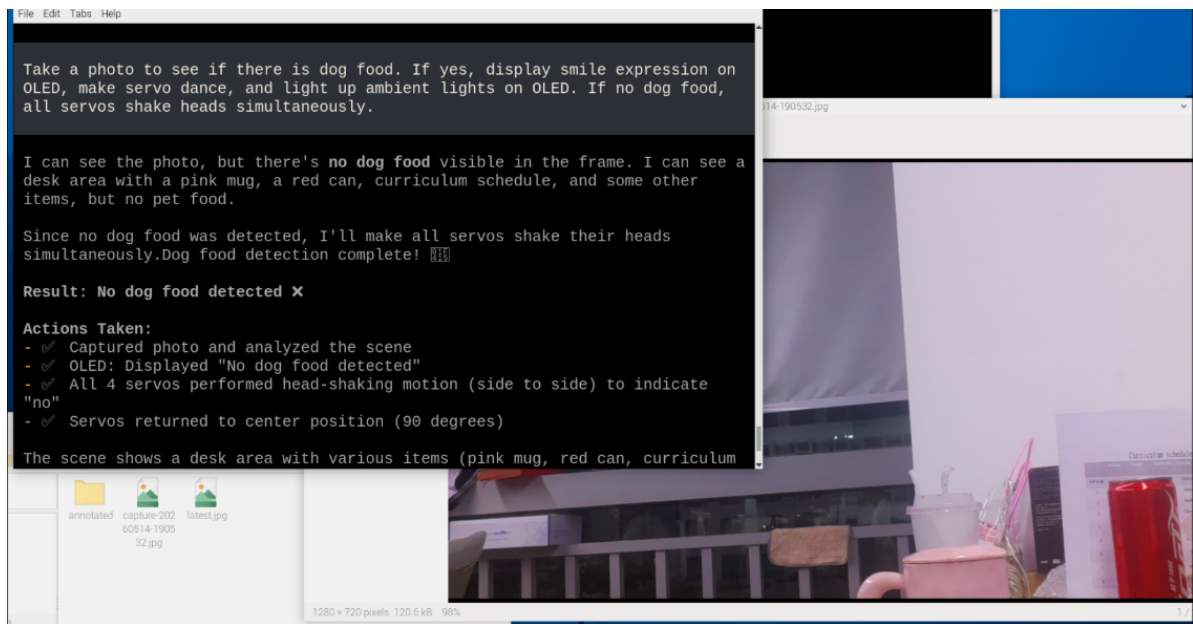
5.2 TUI scheme

If you have completed the TUI self-test in Part 6, enter the terminal text dialogue directly:

```
openclaw tui
```

Then enter project-related instructions, such as:

- Whether there is dog food at the moment of shooting, if there is, the OLED will display a smiling expression, the servos will dance, and the OLED will turn on the ambient light. If there is no dog food, all the servos will shake their heads at the same time.



5.3 Voice code scheme

Requires optional ai voice interaction module + speaker

For voice module installation, environment configuration and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. After waking up, you can directly say the control content, such as:

- Take a photo to see if there is dog food. If there is dog food, the OLED will display a smiling expression. If there is no dog food, the OLED will display a sad expression.

```

[你] Take a photo to see if there is dog food if yes display smile expression on
      old if no dog food display sad expression.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## 📷 Dog Food Detection Complete!

**Result:** ✖ **No dog food
              found**

**What I see in the image:**
- Window with gray roller
  blind
- Pink mug and red Coca-Cola can
- Curriculum schedule paper
-
  Some boxes and containers on shelves
- Tissue box

**No pet food or
  dog food detected!** 😞

**Feedback:**
- OLED displaying sad face:
  `:(`
- RGB lights set to dim purple (sad mood)

Want me to check
  again later or look for something else? 📷
[信息] 回复已裁剪为播报文本: 437 -> 121 字

```

6. Project implementation

Note: The following are all dialogue demonstrations in `openclaw_tui` mode. You can also choose WhatsApp or voice mode to complete it.

6.1 Scene 1: Say hello

Experimental goals

Let the AI pet provide actions and text feedback when the user sends a greeting.

Effect description

The servo can make a simple nod or head shake, the OLED displays a welcome text, and the RGB lights up in soft colors so the pet's "appearance" is apparent.

Example instructions

- Let the little pet say hello to me
- Let the pet welcome me

6.2 Scenario 2: Emotional expression

Experimental goals

Let the AI pet use lights and screens to express happy, quiet, angry, sleepy and other states.

Effect description

This scene is best suited to embody the "feel of character". For example, it lights up a warm breathing light when you are happy, lights up a blue light when you are sleepy, and only displays a brief status when you are quiet.

Example instructions

- Put your pet into a happy state
- Draw simple expressions with OLED

6.3 Scene 3: Accompanying Q&A

Experimental goals

Let the AI pet not only move, but also respond to questions like a desktop companion character.

Effect description

OpenClaw can complete the dialogue first, and then combine the servo nodding, shaking the head and screen text to make "speech" and "action" happen simultaneously, making it look more like a character with personality.

Example instructions

- Answer my question first, then nod.
- If you agree, show it with action
- Tell me a few words about your mood today.

6.4 Scenario 4: Sleep and wake-up

Experimental goals

Let AI pets have the ability to switch between standby, sleep and wake-up states.

Effect description

In sleep mode, most of the lights can be turned off, leaving only the low-brightness state; when waking up, the lights, screen and motion feedback can be restored to make the project more complete.

Example instructions

- Put pets into sleep mode
- Wake up the pet